Where to Measure Next? Evaluating Feedback-Driven Region Selection for Spatial Transcriptomics

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Abstract

Spatial transcriptomics is costly, so many studies profile only selected regions of a section, and recent methods choose each next region from expression already measured. A recent benchmark of active spot selection for training predictors across patients reported inconsistent gains over random sampling; we ask instead whether, within a slide, measured feedback improves selection over a fixed design matched to the objective. We built a retrospective simulator from 22 human HEST-1k slides (18 studies, four tissues), fixed the image encoder and imputer, registered three analysis stages in advance, and drew inference at the study level. **RQ1:** for whole-section reconstruction, a feedback policy targeting the imputer's error was worse than a non-adaptive design chosen on held-out studies at every budget (0.5–20%), by up to 0.092 Pearson correlation; a semi-synthetic control did not establish sensitivity to reconstruction gains. **RQ2:** for capturing heterogeneous regions, informative feedback raised recall over permuted feedback by up to 0.063, but no margin survived multiplicity correction, and a diversity sampler did about as well. **RQ3:** when a coverage design spent half or more of the budget and feedback the rest, no hybrid beat the coverage design; from 2% upward all were within 0.007 correlation and 0.006 recall, although gains near 0.02 and 0.03 would probably have been detected at 2–10%. Feedback was informative (permuted values made the same policies worse) but mainly offset the cost of leaving coverage. **RQ4** (exploratory): batches of five neither eroded nor established a detection advantage. Method developers should compare adaptive selection with objective-matched fixed designs and with itself fed uninformative values. For experimentalists, within simulated spot-level budgets, one imputer, and computational endpoints, no tested feedback policy outperformed a well-chosen fixed design.

1 Introduction

Spatial transcriptomics measures gene expression without dissociating tissue, so expression can be read in its histological context (Ståhl et al., 2016; Rao et al., 2021; Moses & Pachter, 2022). Whole-transcriptome assays remain costly, and region-of-interest platforms such as GeoMx DSP (Merritt et al., 2020) profile only selected regions of a section. Expression elsewhere can be predicted from the co-registered hematoxylin and eosin (H&E) image (Xie et al., 2023; Jaume et al., 2024). Which regions are measured therefore determines both how well the unmeasured tissue can be reconstructed and whether distinct tissue compartments are measured at all.

Choosing regions is an experimental-design problem, and recent methods solve it sequentially. SOFisher (Li et al., 2026) uses reinforcement learning to move fields of view toward regions enriched for a target signal, and SCR^2-ST (Zhu et al., 2026a) learns where to sample spots to build training sets for expression prediction. S2-omics (Yuan et al., 2025) takes the opposite approach and selects regions from histology alone, before any measurement. The appeal of sequential selection follows active learning (Settles, 2009): each measurement can inform the next. Its cost is a measurement turnaround between choices. A recent benchmark of active spot selection for training expression predictors found inconsistent gains over random sampling at small budgets (Zhu et al., 2026b), but addressed a different decision (Section 2). We are not aware of an evaluation

that compares feedback-driven region selection within a slide against fixed designs matched to the objective while ablating the measured feedback itself.

Theory gives different expectations for different goals. For reconstructing a Gaussian-process field with a known kernel, posterior variance does not depend on the observed values, so a design fixed before any measurement is as good as a sequential one; when the model's parameters are uncertain, sequential designs can do better (Krause et al., 2008). Imputers learned from images are not Gaussian processes with known kernels, so for reconstruction the question is empirical. Finding regions defined by their expression is closer to active search, in which each choice exploits the labels already observed to locate further targets (Garnett et al., 2012).

An evaluation that can separate these cases needs four features. The baseline must be a design that could be chosen before the target slide is profiled, because selecting the best design on the evaluation data biases the comparison (Varma & Simon, 2006; Cawley & Talbot, 2010). Generalisation must be assessed across studies, not within them (Bernau et al., 2014), with slides from one study treated as dependent (Field & Welsh, 2007). The contribution of feedback must be isolated by giving the same policy uninformative values, in the spirit of permutation tests for learning algorithms (Ojala & Garriga, 2010). Finally, null results need positive controls and power analysis to be interpretable (Altman & Bland, 1995).

We built such an evaluation and asked four questions. **RQ1**: does conditioning region selection on measured expression improve whole-section reconstruction over a feasible non-adaptive design? **RQ2**: does it improve the capture of molecularly heterogeneous regions? **RQ3**: does feedback add to a strong coverage design when it allocates only the remainder of the budget? **RQ4**, an exploratory stress test: does a detection advantage persist when measurements arrive in batches? The study contributes an evaluation, not a method: a reproducible simulator on 22 human slides from 18 studies, analyses registered before each experimental stage, and evidence that the tested feedback policies offered no supported advantage over well-chosen fixed designs for either objective.

2 Related work

Expression from histology. Expression has been predicted from H&E by contrastive retrieval (Xie et al., 2023), flow matching over spatial neighbourhoods (Huang et al., 2025), diffusion (Zhu et al., 2025), and autoregressive generation (Ouyang et al., 2025), and benchmarked on HEST-1k (Jaume et al., 2024). These methods take the measured regions as given. We hold the predictor fixed and vary only the regions it receives.

Choosing what to measure. Pool-based active learning selects queries by uncertainty (Lewis & Gale, 1994), expected error reduction (Roy & McCallum, 2001), or coverage of the input space (Sener & Savarese, 2018). Sensor placement offers the matching non-adaptive designs: variance- and mutual-information-greedy selection, which are near-optimal for Gaussian-process reconstruction (Krause et al., 2008). Adaptive submodularity extends such guarantees to greedy policies that condition on observations (Golovin & Krause, 2011), and active search addresses the related goal of collecting positives (Garnett et al., 2012). In batch acquisition, the members of a batch are chosen without each other's labels (Kirsch et al., 2019), and global optimisation commonly begins with a space-filling design before switching to adaptive queries (Jones et al., 1998). Active feature acquisition casts measurement as a sequential policy (Yin et al., 2020).

Design in spatial omics. Gene-panel selectors fix which genes to measure (Covert et al., 2023; Lu et al., 2026). Region selectors range from image-based rankers such as SpatialFinder, which score regions from H&E before any expression is measured (Xu et al., 2026), and histology-based heterogeneity selection (Yuan et al., 2025) to sequential reinforcement-learning samplers (Li et al., 2026; Zhu et al., 2026a). Our evaluation includes reimplementations of these objectives and a native run of SOFisher. The closest evaluation is a retrospective benchmark of active spot selection on breast and skin cancer cohorts (Zhu et al., 2026b), which compared uncertainty- and diversity-based strategies with random sampling and found that active strategies did not consistently win at small budgets. Its question differs from ours in three ways. It selects spots to train a morphology-to-expression predictor that is evaluated on held-out patients, whereas we select regions within a slide for a fixed imputer. Its baseline is random sampling, whereas ours are non-adaptive designs

chosen for each objective. It compares selection strategies, whereas we hold each strategy fixed and remove the information in the measured values.

3 Methods

3.1 Acquisition simulator

A slide is a set of regions, each with an image embedding, spatial coordinates, and measured expression. A policy reveals regions one at a time until a budget, a fixed fraction of the slide's regions, is spent; each revealed region exposes its measured expression to the policy. The simulator is retrospective, as in pool-based active-learning evaluation (Settles, 2009): every region has been measured, so any selection can be scored against ground truth. For RQ4, regions are revealed in rounds, and within a round the policy can use only measurements available at its start. Stage 1 also tested contiguous blocks of 7 regions around a chosen seed, approximating field-of-view acquisition. Budgets range from 0.5% of regions (9 to 45 regions depending on slide size, comparable to ROI-based profiling) to 20%.

3.2 Slides and preprocessing

We used 22 human HEST-1k slides (Jaume et al., 2024) from prostate (6), kidney (4), lung (6), and colon (6), of which 7 are Visium HD (de Oliveira et al., 2025) and 15 standard Visium (Ståhl et al., 2016) (Appendix A). The tissues are the four used to evaluate the image-based ranker we compare against (Xu et al., 2026). Selection rules were fixed before any result on these slides: human tissue, whole-transcriptome assay, at least 1,000 spots under tissue, a source image of at most 0.6 μm/px so that image patches are not substantially upsampled, one slide per specimen, and at most two slides per study. The slides come from 18 studies, which serve as the independent units of inference.

A region is one native spot, the unit at which HEST provides both expression and an image patch. Visium spots are 55 μm wide on a 100 μm pitch; the HD pseudo-spots distributed by HEST have a median spacing of 128 μm on our slides. Slides contain 1,766 to 9,080 regions. Each region's image input is the HEST 224 px patch at 0.5 μm/px (Jaume et al., 2024), embedded with UNI2-h, a later release of the pathology foundation model UNI (Chen et al., 2024),[1] used frozen. Following the HEST benchmark, expression is log1p-transformed and genes detected in fewer than 10% of regions are removed (Jaume et al., 2024).

3.3 Objectives

Reconstruction (RQ1, RQ3): a fixed imputer predicts the unprofiled regions, scored by the per-gene Pearson correlation over those regions averaged over the 50 most variable genes (PCC), the HEST benchmark target (Jaume et al., 2024). *Detection* (RQ2–RQ4): regions of interest (ROIs) are the 20% of regions with the highest local molecular heterogeneity, which we define as the variance of top-50-gene expression across each region's 8 nearest spatial neighbours. The definition is ours; it is motivated by the goal of capturing molecular heterogeneity that drives S2-omics (Yuan et al., 2025), which itself selects regions from histology. The ROI fraction equals the largest budget, so a perfect policy could capture every ROI. We score recall, the fraction of ROIs profiled. Gene sets and ROIs are computed from ground truth and used only for scoring.

3.4 Imputer

One imputer serves every policy, so differences between policies reflect selection alone. It maps image principal components through random Fourier features (Rahimi & Recht, 2007), fits a ridge regression to the profiled regions, and propagates the resulting predictions over a fixed 8-nearest-neighbour graph in image and spatial features while holding profiled regions at their measured values, as in harmonic label propagation (Zhu et al., 2003). This is a light analogue of neighbourhood-aware predictors such as STFlow (Huang et al., 2025), cheap enough to refit after every measurement. Its predictions depend on the measured values. Its uncertainty, one minus the graph influence that reaches a region from profiled nodes, depends only on which

[1] https://huggingface.co/MahmoodLab/UNI2-h

regions are profiled. A k-nearest-neighbour imputer and a regression-kriging imputer (Hengl et al., 2007) were evaluated on an earlier slide set (Appendix C). Hyperparameters of the imputer and policies were fixed before the registered analysis in which each was first used and were not tuned on its outcomes; those of the imputer were set during development on one prostate slide (Appendix B).

3.5 Policies

Non-adaptive designs. Random; farthest-point selection in space (grid), a greedy k-centre cover (Gonzalez, 1985); k-means (Lloyd, 1982) on image embeddings, taking the region nearest each centroid; a one-shot ranker after SpatialFinder (Xu et al., 2026), which scores regions once from image embeddings by rare-cluster membership and distance to the cluster centroid, a simplification of its diversity score without the cell classifier or pathologist labelling ("one-shot" below); and variance- and mutual-information (MI)-greedy placement (Krause et al., 2008) with an RBF kernel on image and coordinate features and a median-heuristic lengthscale (Garreau et al., 2017).

Sequential fixed designs. Error-reduction (Roy & McCallum, 2001) picks the region whose 15 nearest unprofiled neighbours in the imputer's feature space carry the most uncertainty, and maximum-uncertainty picks the most uncertain region (Lewis & Gale, 1994). Because the imputer's uncertainty does not depend on measured values, both are fixed designs computed one region at a time; for error-reduction we confirmed that permuted values leave its selections unchanged.

Value-conditioned policies. The *residual* policy keeps the error-reduction form but replaces the uncertainty with an estimate of the imputer's actual error. It computes the leave-one-out residual of the image ridge model at each profiled region from the hat matrix (Hastie et al., 2009), averages it over each unprofiled region's 10 nearest profiled neighbours, and multiplies by one minus the graph influence. The *local-variance* policy, our detection policy, scores a candidate by the kernel-weighted variance of measured expression around it plus 0.5 times its distance to the nearest profiled region, an exploitation term for heterogeneity balanced by an exploration term as in active search (Garnett et al., 2012).

Competitor objectives. A diversity sampler applies farthest-point selection in joint image and spatial features, the coverage criterion of core-set active learning (Sener & Savarese, 2018). It approximates the coverage and diversity terms of the SCR^2-ST reward, which rewards covering single-cell states, cell-type diversity, and spatial spread (Zhu et al., 2026a). An adaptive ROI-seeking sampler follows the reward of SOFisher (Li et al., 2026). Both reimplement stated objectives, not the authors' trained models.

Hybrids (RQ3). A hybrid takes its first $\lceil fn \rceil$ regions, for a budget of n regions and $f \in \{0.5, 0.8\}$, from a fixed coverage design and the rest from a feedback policy: the residual policy for reconstruction and local-variance for detection. This mirrors the space-filling start of adaptive global optimisation (Jones et al., 1998). Because the coverage designs select in a fixed order, the first phase is exactly the prefix of the design's own selection (verified for all four designs at both fractions), so a hybrid and its coverage design differ only in how the last $1 - f$ of the budget is spent.

3.6 Feedback ablations

Each value-conditioned policy and hybrid was also run with uninformative feedback while its selection rule was left unchanged. Under *permuted* feedback, profiling a region reveals the expression of a randomly chosen other region, so the residual policy's leave-one-out model, for example, is fitted to permuted values. Under *constant* feedback, every region appears to have the slide's mean expression. The imputer and the scores always use the true values. A policy that gains from feedback should outperform its own permuted version, the logic of permutation tests for whether a learner exploits real structure (Ojala & Garriga, 2010). Stage 2 used one permutation; stage 3 used two per configuration and repeated the stage-2 ablations with up to four further seeds.

3.7 Comparators

The primary stage-2 comparator is a *feasible* design chosen without the target slide. For each held-out study, the non-adaptive design (grid, k-means, variance-greedy, MI-greedy, or one-shot) with the best mean score on the other 17 studies at that budget and objective is applied to the held-out slides, a leave-one-study-out form of cross-study validation (Bernau et al., 2014). The best design for each slide chosen in hindsight is reported only as a benchmark, since selecting on the evaluation data inflates the comparator (Varma & Simon, 2006; Cawley & Talbot, 2010). Stage 3 compared each hybrid with a single design fixed for all slides at each budget and objective: the design the feasible procedure chose for most held-out studies. Those designs were chosen on stage-2 results from the same slides, which favours the baseline.

3.8 Statistical analysis and registration

Each margin is a per-slide difference between two policies. We report its mean with a 95% percentile bootstrap interval (Efron & Tibshirani, 1993) that resamples whole studies (Field & Welsh, 2007; Cameron et al., 2008) (5,000 resamples), and an exact two-sided sign-flip test over the 2^{18} study-level sign patterns (Winkler et al., 2014). Intervals are not adjusted for multiplicity; p_H denotes p-values adjusted by Holm's procedure within a registered family (Holm, 1979). To make null results interpretable (Altman & Bland, 1995), we report the minimum detectable effect (Bloom, 1995): the smallest true mean margin detected with 80% power (Cohen, 1988), estimated by simulation that resamples studies around the observed dispersion (Arnold et al., 2011). For a family of m margins we compute it at $\alpha = 0.05/m$ (MDE_H), a conservative bound for the Holm decision.

The sign-flip test assumes exchangeable study margins. This holds for a fixed comparator but not exactly for the feasible one, whose selection for each study uses the other 17. Following the principle that model selection must be repeated inside resampling (Varma & Simon, 2006), we checked the stage-2 results post hoc with a bootstrap that repeats the selection in every replicate and with a single fixed comparator (Appendix F).

Analyses were registered in the project's version-controlled repository in three stages (Nosek et al., 2018), each before its experiments and after the previous stage's results were known.[2] Stage 1 compared policies with hindsight benchmarks in a family of 24 margins, including the first batched experiments (Appendix D). Stage 2 introduced the feasible comparator, the residual policy, and the feedback ablations, with a primary family of 18 margins: residual minus feasible at six budgets (RQ1), and at four budgets local-variance minus its permuted version, minus the feasible design, and minus the diversity sampler (RQ2). Stage 3 tested the hybrids, with a primary family of 20 margins, each hybrid minus the same coverage design run for the whole budget (RQ3), and a secondary family of 24 that compared hybrids with their permuted versions and repeated the batched experiments with the fixed detection design and permuted controls (RQ4). In every stage a benefit was to be claimed only for a positive margin with $p_H < 0.05$. Results are labelled confirmatory (registered primary), secondary (registered, Holm within the secondary family), descriptive, or post hoc.

3.9 Positive controls

A null result is informative only if the protocol could have registered an effect. In synthetic slides of 20×20 regions with spatially clustered structure that the image cannot predict (8 slides per setting, 10% budget, k-NN imputer), we varied the correlation length of the hidden structure and the amplitude α with which ROIs appear in measured expression. On the 22 real slides we injected 20 genes carrying three spatial hotspots of radius 5% of the slide diagonal, in the manner of simulated spatially variable genes (Svensson et al., 2018), with standard deviation $\alpha \in \{0, 1, 2\}$ times the median of the slide's top-50 genes. Because the image cannot predict the hotspots, only measured values reveal them. Before running this control we set its success criterion: the residual policy beats the feasible design with an interval excluding zero at $\alpha = 2$.

[2]Verbatim excerpts of the three plans and their commit dates are in the supplementary material (`registered_plans/`); the plans label RQ4 as Q3.

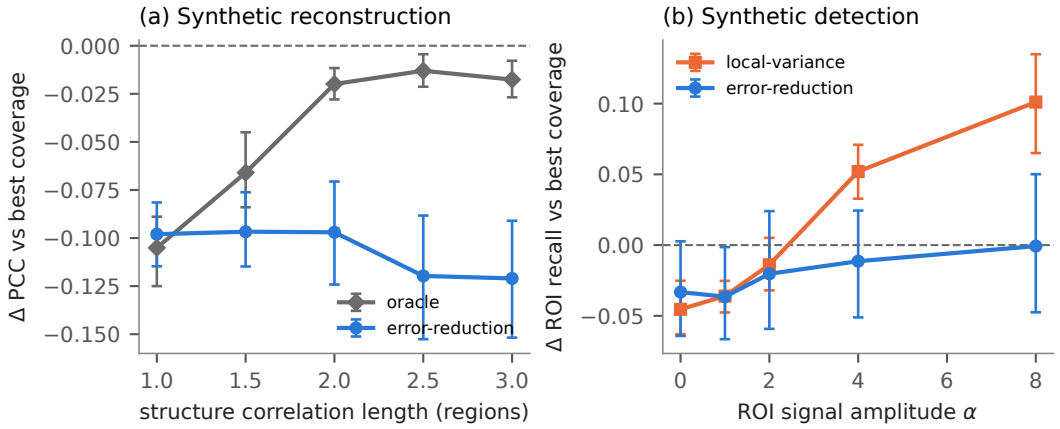


Figure 1: Synthetic controls (8 slides per point, 10% budget, mean and 95% bootstrap CI over slides). (a) Reconstruction margin over the best coverage design versus the correlation length of the hidden structure, for an ordering by true local difficulty and for error-reduction. (b) Detection-recall margin versus ROI amplitude.

4 Results

4.1 Positive controls

The simulator registered an effect of feedback on detection. In synthetic slides, the local-variance policy's recall margin over coverage rose with ROI amplitude, from -0.045 at $\alpha = 0$ to $+0.052$ (95% CI $+0.033$ to $+0.071$) at $\alpha = 4$ and $+0.101$ ($+0.065$ to $+0.135$) at $\alpha = 8$, positive on all 8 slides at both amplitudes (Figure 1b). Reconstruction behaved differently: even an ordering by true local difficulty, which no policy could deploy, stayed below coverage at every correlation length, by 0.105 and 0.066 PCC at the two shortest and 0.013 to 0.020 at the longer ones (Figure 1a).

The semi-synthetic control failed its criterion. At $\alpha = 1$ and 2, all 20 injected genes entered the evaluation set on every slide, yet at $\alpha = 2$ the residual policy remained below the feasible design, by 0.021 (0.012 to 0.030) PCC at 5% and 0.011 (0.004 to 0.019) at 10%, close to its deficits without injection (0.015 and 0.007; Figure 2c, Appendix H). Feedback still helped relative to permuted values, by $+0.001$ to $+0.013$. The protocol can thus register a large detection effect, but not demonstrably a reconstruction gain, and the RQ1 results below concern the tested policies.

4.2 RQ1: Reconstruction

No value-conditioned policy outperformed the feasible design (Tables 1 and 2, Figure 2a). The residual policy fell below it at every budget, by 0.092 PCC at 0.5%, 0.057 at 1%, 0.025 at 2%, 0.016 at 5%, 0.007 at 10%, and 0.002 at 20%, significantly so at 0.5, 1, 5, and 10% ($p_H \leq 0.010$). The local-variance policy was further below, by 0.119 to 0.163 (secondary, $p_H \leq 0.001$). Bootstrap intervals that repeat the comparator's selection in each replicate excluded zero at every budget (post hoc; Appendix F).

The measured values nonetheless carried information. With informative rather than permuted feedback, the residual policy gained 0.031, 0.039, and 0.030 PCC at 0.5–2% (secondary, Appendix G; p_H 0.056, 0.002, 0.014), with no difference at 10–20%. Feedback moved the policy toward the feasible design without closing the gap. The error-reduction design, which shares the residual policy's structure but uses value-independent uncertainty, trailed the feasible design by similar amounts (0.107, 0.067, and 0.024 at 0.5–2%; post hoc), and both converged on it as the budget grew. The only design to exceed the feasible design was maximum-uncertainty, by 0.003 PCC at 20% (95% CI 0.002 to 0.004, $p = 0.0007$, post hoc), a fixed design that uses no feedback. The hindsight benchmark exceeded the feasible design by 0.016 to 0.021 PCC at budgets up to 2% and by at most 0.002 above (Appendix E).

Table 1: Reconstruction accuracy (PCC over the top-50 genes on unprofiled regions), mean over 22 slides. "Feasible" is the non-adaptive design chosen on the other studies; "per-slide best" is chosen in hindsight. "Residual, perm." is the residual policy given permuted feedback.

	Non-adaptive		Sequential fixed design		Value-conditioned		
Budget	feasible	per-slide best	error-red.	max-unc.	residual	residual, perm.	local-var.
0.5%	0.449	0.470	0.343	0.348	0.357	0.327	0.348
1%	0.534	0.550	0.467	0.462	0.477	0.438	0.424
2%	0.591	0.607	0.567	0.562	0.566	0.536	0.471
5%	0.659	0.660	0.650	0.648	0.643	0.633	0.513
10%	0.685	0.687	0.684	0.685	0.678	0.677	0.535
20%	0.707	0.709	0.708	0.710	0.705	0.705	0.544

Table 2: Reconstruction margins (PCC), 22 slides, 18 studies. Δ: mean per-slide difference with an unadjusted 95% CI resampling studies. p_H: Holm-adjusted within the stage-2 primary or secondary family; p: unadjusted (post hoc).

	residual − feasible (primary)		residual − residual, perm. (secondary)		error-red. − feasible (post hoc)	
Budget	Δ [95% CI]	p_H	Δ [95% CI]	p_H	Δ [95% CI]	p
0.5%	−0.092 [−0.132, −0.054]	0.002	+0.031 [+0.010, +0.057]	0.056	−0.107 [−0.149, −0.067]	<0.001
1%	−0.057 [−0.089, −0.028]	0.010	+0.039 [+0.021, +0.060]	0.002	−0.067 [−0.093, −0.044]	<0.001
2%	−0.025 [−0.041, −0.009]	0.099	+0.030 [+0.014, +0.049]	0.014	−0.024 [−0.039, −0.007]	0.013
5%	−0.016 [−0.025, −0.009]	0.003	+0.009 [+0.001, +0.018]	0.238	−0.010 [−0.017, −0.003]	0.021
10%	−0.007 [−0.011, −0.003]	0.010	+0.001 [−0.002, +0.004]	1.000	−0.001 [−0.006, +0.003]	0.763
20%	−0.002 [−0.004, −0.001]	0.126	−0.001 [−0.003, +0.002]	1.000	+0.001 [−0.002, +0.003]	0.537

4.3 RQ2: Detection

Informative feedback raised the local-variance policy's recall over permuted feedback by 0.005, 0.017, 0.029, and 0.063 at 2, 5, 10, and 20% (Table 3, Figure 2b). The unadjusted intervals excluded zero at 5–20%, but no margin survived adjustment ($p_H \geq 0.29$), and each estimate lay below its family-wise MDE (0.015, 0.035, 0.066, 0.112). The margin over the feasible design was smaller (0.006 to 0.026, $p_H \geq 0.25$), and the margin over the diversity sampler was close to zero (0.001 to 0.012, $p_H = 1.000$).

The absolute recalls locate the feedback effect (Table 4). At 20%, the policy recovered 0.278 of the ROIs with informative feedback but only 0.215 with permuted and 0.204 with constant values, below the feasible design (0.252); the diversity sampler reached 0.266 without feedback. Part of the feedback effect therefore recovers ground that the selection rule loses on its own. The margin over constant feedback at 20% was the only significant detection effect of feedback (+0.074, $p_H = 0.038$, secondary).

The apparent advantage also depends on the comparator. Against MI-greedy fixed for every slide, the local-variance policy gained 0.043 and 0.078 recall at 10 and 20% ($p_H = 0.043$ for both; post hoc). MI-greedy, however, is a reconstruction design and the weakest detection design in the pool at 5–20% (0.200 recall at 20%, against 0.247 for variance-greedy and 0.252 for the one-shot ranker), and selection-aware intervals against the feasible design included zero at every budget (Appendix F). The ROI-seeking port did not improve on coverage in stage 1 (Appendix D). Because the port is not the trained SOFisher, we also ran SOFisher natively on its mouse cortex data, where its reward for captured target regions rose from a median of −238 early in training to +147 late (descriptive; Appendix L); the port's result is therefore not a test of SOFisher. SCR^2-ST could not be run natively because its processed training data were not public when we ran our experiments.

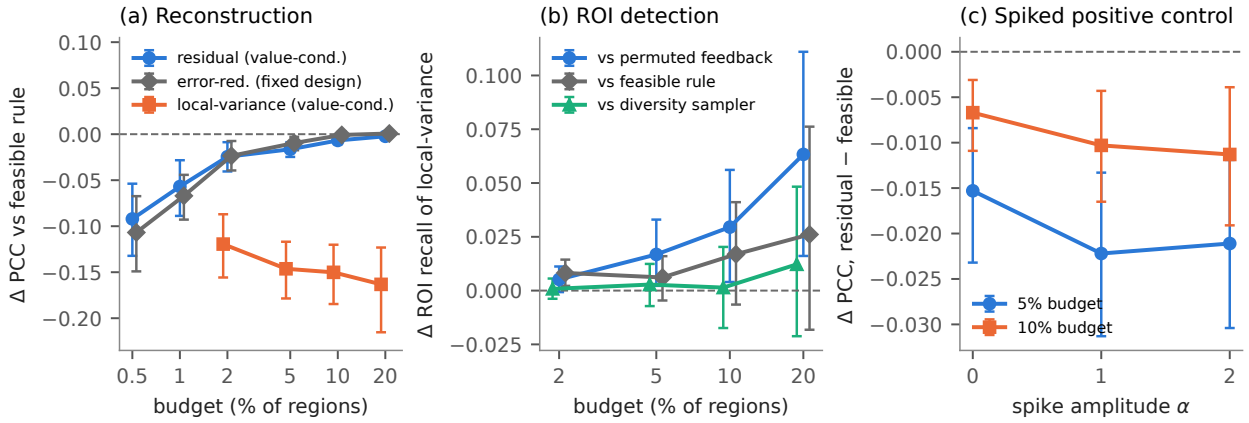


Figure 2: Stage-2 margins over 22 slides, mean and unadjusted 95% CI resampling studies. (a) Reconstruction margin over the feasible design. (b) Detection-recall margin of the local-variance policy over its permuted-feedback version, the feasible design, and the diversity sampler. (c) Spiked control: residual policy minus feasible design by spike amplitude.

Table 3: Detection margins (ROI recall) of the local-variance policy, 22 slides. Δ with unadjusted 95% CI resampling studies; p_H Holm-adjusted over the 18 stage-2 primary margins; MDE_H at 80% power and $\alpha = 0.05/18$.

	local-var. − permuted feedback			local-var. − feasible		local-var. − diversity	
Budget	Δ [95% CI]	p_H	MDE_H	Δ [95% CI]	p_H	Δ [95% CI]	p_H
2%	+0.005 [−0.001, +0.011]	1.000	0.015	+0.008 [+0.002, +0.014]	0.250	+0.001 [−0.004, +0.006]	1.000
5%	+0.017 [+0.002, +0.033]	0.530	0.035	+0.006 [−0.005, +0.016]	1.000	+0.003 [−0.007, +0.012]	1.000
10%	+0.029 [+0.004, +0.056]	0.530	0.066	+0.017 [−0.006, +0.041]	1.000	+0.001 [−0.017, +0.020]	1.000
20%	+0.063 [+0.016, +0.111]	0.294	0.112	+0.026 [−0.018, +0.076]	1.000	+0.012 [−0.021, +0.048]	1.000

4.4 RQ3: Feedback on top of a coverage design

No hybrid outperformed its coverage design (Table 5, Figure 3). For reconstruction, the hybrids recovered most of the residual policy's deficit, but the half-coverage hybrid remained below k-means at 0.5%, by 0.036 PCC ($p_H = 0.006$), and from 2% upward every hybrid was within 0.007 PCC of MI-greedy with no significant margin. For detection, every hybrid margin lay between −0.003 and +0.006 recall ($p_H \geq 0.76$). These nulls are tight: at 2–10% the family-wise MDEs were 0.004 to 0.017 PCC and 0.004 to 0.026 recall, so gains above about 0.02 PCC or 0.03 recall would probably have been detected. At 20% the detection bound widens to 0.035–0.068.

Feedback within the hybrids was informative. With permuted values the hybrids did worse at 2%, by 0.016 and 0.007 PCC for the two reconstruction hybrids and 0.007 recall for the half-coverage detection hybrid (secondary, $p_H \leq 0.027$). For that detection hybrid the difference grew to 0.014–0.035 at 5–20% without surviving correction; for reconstruction at 10–20% the permuted versions were marginally ahead, by at most 0.003 PCC ($p_H \geq 0.12$; Appendix I). The absolute scores show what the information bought (Table 16). With permuted values the feedback phase lost ground relative to continuing the coverage design, for example from 0.127 to 0.110 recall at 10%, and real values restored the hybrid to about the level of the design (0.133). In detection, the local-variance policy run alone (0.137 at 10%, 0.278 at 20%) scored above both hybrids at every budget (descriptive).

4.5 RQ4: Batched and contiguous acquisition (exploratory)

In stage 1, at 10% budget, the local-variance policy's detection margins over the hindsight benchmark were +0.021, +0.034, and +0.011 for rounds of 1, 5, and 25 regions and +0.024 for contiguous blocks, none with

Table 4: ROI-detection recall, mean over 22 slides. "Observed", "permuted", and "constant" denote the local-variance policy with informative, permuted, and constant feedback. The oracle profiles regions in order of true heterogeneity and bounds recall from above.

	Non-adaptive				Value-conditioned (local variance)			
Budget	feasible	k-means	MI-greedy	diversity	observed	permuted	constant	oracle
2%	0.018	0.018	0.018	0.026	0.027	0.021	0.022	0.100
5%	0.064	0.050	0.046	0.067	0.070	0.053	0.051	0.250
10%	0.120	0.110	0.094	0.136	0.137	0.107	0.103	0.500
20%	0.252	0.232	0.200	0.266	0.278	0.215	0.204	0.999

Table 5: Stage-3 primary margins: hybrid minus the same fixed coverage design run for the whole budget, 22 slides. The hybrid takes the first 50% or 80% of regions from the design and the rest from the feedback policy. Δ with unadjusted 95% CI resampling studies; p_H Holm-adjusted over the 20 primary margins; MDE_H at 80% power and $\alpha = 0.05/20$.

		coverage for 50% of budget			coverage for 80% of budget		
Budget	fixed rule	Δ [95% CI]	p_H	MDE_H	Δ [95% CI]	p_H	MDE_H
Reconstruction (PCC); feedback: residual policy							
0.5%	k-means	-0.036 $[-0.057, -0.018]$	0.006	0.056	-0.009 $[-0.021, +0.003]$	1.000	0.033
1%	k-means	-0.010 $[-0.020, +0.002]$	1.000	0.021	-0.009 $[-0.019, -0.001]$	1.000	0.025
2%	MI-greedy	$+0.001$ $[-0.007, +0.011]$	1.000	0.017	$+0.007$ $[+0.002, +0.013]$	0.359	0.013
5%	MI-greedy	-0.003 $[-0.007, +0.002]$	1.000	0.013	-0.001 $[-0.004, +0.003]$	1.000	0.008
10%	MI-greedy	0.000 $[-0.002, +0.002]$	1.000	0.005	$+0.001$ $[-0.001, +0.002]$	1.000	0.004
20%	MI-greedy	-0.002 $[-0.004, +0.000]$	1.000	0.005	-0.001 $[-0.003, +0.001]$	1.000	0.006
Detection (ROI recall); feedback: local-variance policy							
2%	variance-greedy	$+0.003$ $[+0.000, +0.005]$	0.756	0.007	0.000 $[-0.001, +0.002]$	1.000	0.004
5%	variance-greedy	$+0.002$ $[-0.007, +0.010]$	1.000	0.020	0.000 $[-0.003, +0.003]$	1.000	0.008
10%	variance-greedy	$+0.006$ $[-0.005, +0.018]$	1.000	0.026	$+0.001$ $[-0.005, +0.007]$	1.000	0.015
20%	one-shot	0.000 $[-0.033, +0.034]$	1.000	0.068	-0.003 $[-0.019, +0.014]$	1.000	0.035

$p_H < 0.05$ (Appendix D). With a single round the policy has no measurements and falls back to its starting rule, the regions nearest the slide centroid; the resulting margin of -0.051 ($p_H = 0.008$) reflects that rule rather than adaptivity. Reconstruction margins were negative in every setting.

In stage 3, rounds of 5 regions did not reduce recall: the policy captured 0.150 and 0.288 of ROIs at 10 and 20%, slightly more than when choosing one region at a time (0.137 and 0.278; Table 6). Its margins over the fixed detection design ($+0.023$, $+0.035$) and over permuted feedback ($+0.050$, $+0.084$) did not survive correction ($p_H \geq 0.11$).

4.6 Robustness

Permutation seeds. Repeating the stage-2 ablations with further seeds (descriptive; Appendix I) did not change any conclusion. Across seeds the mean margin varied with a standard deviation of at most 0.007 PCC and 0.013 recall. The seed-averaged detection margins ($+0.005$, $+0.017$, $+0.033$, $+0.074$ at 2–20%) stayed below their family-wise MDEs, and for reconstruction the registered seed gave the largest of three margins at 0.5% ($+0.031$, against $+0.024$ and $+0.016$).

Evaluation genes. Choosing the top-50 genes from the whole slide uses information a prospective study would lack. Choosing them from a random 15% reference subset changed the stage-1 reconstruction margins by at most 0.003, and scoring all filtered genes left them negative (-0.033 to -0.079; Appendix J).

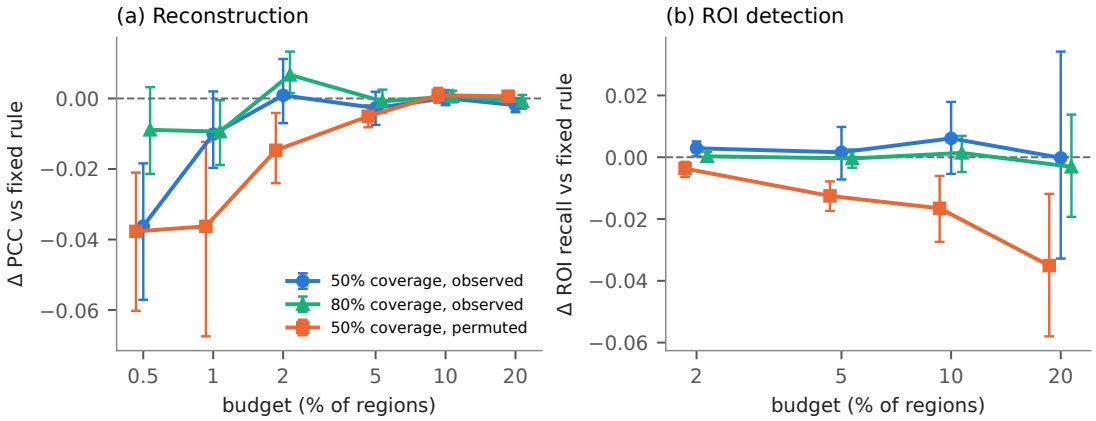


Figure 3: Hybrid minus the same fixed coverage design, 22 slides, mean and unadjusted 95% CI resampling studies. "Observed" hybrids receive real feedback (primary); the permuted series is the half-coverage hybrid given permuted feedback, averaged over two seeds (descriptive). Fixed designs as in Table 5.

Table 6: Batched acquisition, stage 3: local-variance policy revealing regions in rounds of 5, 22 slides. "One at a time" is the stage-2 sequential run. Δ with unadjusted 95% CI resampling studies; p_H Holm-adjusted over the 24 stage-3 secondary margins; permuted values averaged over two seeds.

		ROI recall				rounds of 5 − fixed rule		rounds of 5 − permuted	
Budget	fixed rule	fixed	one at a time	rounds of 5	rounds of 5, perm.	Δ [95% CI]	p_H	Δ [95% CI]	p_H
10%	variance-greedy	0.127	0.137	0.150	0.100	+0.023 [−0.000, +0.046]	0.749	+0.050 [+0.022, +0.075]	0.113
20%	one-shot	0.252	0.278	0.288	0.204	+0.035 [−0.017, +0.095]	1.000	+0.084 [+0.025, +0.141]	0.240

Platform. In an exploratory split, stage-1 detection margins were larger on Visium HD than on Visium (+0.048 against +0.008 at 10%), but with five HD studies the smallest attainable p is 0.0625, and platform is confounded with study (Appendix K).

5 Discussion

Reconstruction. The measured values were informative, yet using them did not help. Real feedback beat permuted feedback by up to 0.039 PCC when it drove the whole selection and by up to 0.026 when it chose only the remainder, but in both cases the gain mainly offset the cost of departing from a coverage design. This pattern is what sensor-placement theory leads one to expect when uncertainty depends on locations alone (Krause et al., 2008). Our imputer's uncertainty has this property while its predictions do not, so any benefit of feedback had to come through the predictions. The evaluation does not test the theory, and the failed semi-synthetic control means we cannot exclude value-conditioned policies that would help; it shows that the obvious ones do not.

Detection. Here the answer depends on the baseline. Feedback-driven selection outperformed a design built for reconstruction (MI-greedy, post hoc) and, at the largest budget, its own constant-feedback version. Such gains are what active-search reasoning predicts (Garnett et al., 2012), and the synthetic control confirmed that the simulator can show them. Against a design chosen for detection or a diversity sampler, the margins were small and uncertain, and giving the detection design a coverage phase within the policy did not help. The diversity sampler's near-parity suggests that, on these slides, heterogeneous regions are spread widely enough that broad coverage finds many of them. Resolving a difference of the size observed would need more independent studies than the 18 used here, especially at 20%.

Implications. For developers of adaptive acquisition, two comparisons should be standard: a fixed design chosen on other data for the same objective, and the method itself fed uninformative values. Neither

comparison requires new data, and together they separate the value of feedback from the value of a good selection rule. Inference should treat studies, not slides, as the unit. For experimentalists, our evidence is confined to simulated spot-level acquisition at 0.5–20% of regions, one imputer, and computationally defined endpoints. Within those limits, no tested feedback policy outperformed a fixed design: k-means at the smallest budgets and MI-greedy above them for reconstruction, and variance-greedy, the one-shot ranker, or a diversity design for heterogeneity. Fixed designs also avoid the measurement turnaround that sequential selection requires. Whether this holds for ROI-scale acquisition, other predictors, or annotated biological targets would need prospective or annotated data. Our findings are consistent with the benchmark's result for predictor training (Zhu et al., 2026b), although the two studies address different decisions.

Limitations. The simulator is retrospective: a revealed region returns its measured expression without added noise, batch effects, or delay, conditions that if anything favour feedback. The ROIs are the 20% of regions with the highest local-variance score, not annotated structures. High local variance can mark tissue boundaries, mixtures of cell types, or technical noise as well as targets such as tumour margins or immune niches, so the detection results concern computational heterogeneity and do not show whether feedback helps find a particular biological target. The acquisition unit is a spot or HD pseudo-spot, smaller than GeoMx ROIs, and the two units differ in spacing. The batch and block simulations approximate but do not validate any platform. The 22-slide results use one imputer; the k-NN and kriging comparisons rest on five slides. The competitor samplers reimplement objectives; SOFisher was run natively only on its own data and SCR^2-ST not at all. The registered stage-2 ablation tests use one permutation, with further seeds descriptive. The analysis plans were registered in our own version-controlled repository rather than with an external registry, so their timing rests on commit records, which the supplementary material lists. The feasible comparator, the residual policy, the ablations, and the hybrids were added after earlier results were known, although each was registered before its experiments. Budgets above 20% were not tested.

6 Conclusion

Across 22 human slides from 18 studies, conditioning region selection on measured expression did not improve whole-section reconstruction over a well-chosen fixed design, whether feedback drove the whole selection or only its remainder. For capturing heterogeneous regions, feedback helped relative to a design built for the other objective, but not detectably relative to a design built for detection. The value of feedback in region selection is therefore an empirical question for each objective, and it should be answered against feasible, objective-matched baselines and with uninformative-feedback controls.

Reproducibility. The supplementary material contains the code, configuration, result files, the registered analysis plans with their commit dates, and the scripts that regenerate every table and figure from the result files. A public repository will be linked on acceptance. HEST-1k and UNI2-h are available from their providers under their own access terms.

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A Slides

Table 7 lists the slides. "Cluster" indexes the HEST study used as the resampling unit, and "Pitch" is the median nearest-neighbour spacing of regions.

Table 7: The 22 slides (HEST-1k v1.3 metadata).

Tissue	HEST ID	Platform	Spots	μm/px	Pitch (μm)	Cluster	Study (HEST dataset title)
prostate	TENX174	Visium HD	2385	0.274	128	1	Visium HD Spatial Gene Expression Library, Human Prostate ...
prostate	TENX40	Visium	4371	0.345	100	2	Human Prostate Cancer, Adenocarcinoma with Invasive Carcin...
prostate	MEND160	Visium	4079	0.573	100	3	Spatially resolved clonal copy number alterations in benig...
prostate	TENX46	Visium	3043	0.344	100	4	Human Prostate Cancer, Acinar Cell Carcinoma (FFPE)
prostate	TENX41	Visium	2543	0.345	100	5	Normal Human Prostate (FFPE)
prostate	MEND144	Visium	4074	0.341	100	3	Spatially resolved clonal copy number alterations in benig...
kidney	TENX173	Visium HD	2684	0.274	128	6	Visium HD Spatial Gene Expression Library, Human Kidney (F...
kidney	TENX71	Visium	5936	0.274	100	7	Human Kidney, 11 mm Capture Area (FFPE)
kidney	INT21	Visium	4975	0.457	100	8	Tertiary lymphoid structures generate and propagate anti-t...
kidney	INT19	Visium	4948	0.457	100	8	Tertiary lymphoid structures generate and propagate anti-t...
lung	TENX171	Visium HD	1766	0.274	128	9	Visium HD Spatial Gene Expression Libraries, Post-Xenium, ...
lung	TENX170	Visium HD	1877	0.274	128	9	Visium HD Spatial Gene Expression Libraries, Post-Xenium, ...
lung	TENX163	Visium HD	2487	0.274	128	10	Visium HD Spatial Gene Expression Library, Human Lung Canc...
lung	TENX72	Visium	6195	0.274	100	11	Human Lung Cancer, 11 mm Capture Area (FFPE)
lung	TENX62	Visium	3858	0.273	100	12	Human Lung Cancer (FFPE)
lung	MISC20	Visium	3316	0.438	100	13	A spatially resolved atlas of the human lung characterizes...
colon	TENX156	Visium HD	2179	0.274	128	14	Characterization of immune cell populations in the tumor m...
colon	TENX155	Visium HD	2281	0.274	128	14	Characterization of immune cell populations in the tumor m...
colon	TENX70	Visium	9080	0.274	100	15	Human Colorectal Cancer, 11 mm Capture Area (FFPE)
colon	TENX92	Visium	6518	0.274	100	16	Visium CytAssist Gene Expression Libraries of Post-Xenium ...
colon	MISC47	Visium	4818	0.276	100	17	COLON MAP: Colon Molecular Atlas Project
colon	ZEN48	Visium	2385	0.459	100	18	Charting the Heterogeneity of Colorectal Cancer Consensus ...

Changes from an earlier seven-slide analysis. An earlier analysis used seven Visium HD slides and described regions as 112 μm squares. The current HEST metadata lists two of those slides as non-human (TENX186, *Mus musculus*; TENX175, mixed human and mouse), and both are excluded here. The loader had also assumed 0.5 μm/px, whereas the HD images are about 0.274 μm/px, so those regions were about 61 μm squares containing one pseudo-spot each. On the five retained slides, per-slide local-variance margins (reconstruction and detection, four budgets) differ between the two analyses by a median of 0.015 (maximum 0.096); 3 of 40 change sign, two by less than 0.005 and one detection margin from -0.050 to $+0.015$. With seven slides the smallest attainable exact p-value is 0.016, so no earlier margin could have survived correction for 24 comparisons.

B Configuration and implementation checks

All values are fixed in the released `config.py`. The imputer's hyperparameters were set during development on one prostate slide (TENX174, one of the 22) before any registered analysis and were not changed afterwards; the policies' hyperparameters were fixed before the stage in which each policy first appears.

Seeds: global 2021; random policy $\{0, \ldots, 4\}$; the stage-2 permutation uses the global seed, and stage 3 uses seeds 1 and 2 (1–4 for the repeated local-variance ablation). Imputer: image PCA 64, random Fourier features 512 (median-heuristic bandwidth), ridge 10, graph $k = 8$, 20 propagation steps, step size 0.5. k-NN imputer: $k = 10$, joint image and space features. Kriging: image PCA 64, ridge 10, residual $k = 10$. Placement kernels: feature PCA 64, median-heuristic lengthscale, noise 10^{-2}. Error-reduction and residual policies: neighbourhood 15; the residual policy uses the 256 genes with the highest variance among profiled regions and 10 profiled neighbours. Local-variance: spatial kernel width 0.1 of the slide diagonal, exploration weight 0.5. Hybrids: coverage fraction $f \in \{0.5, 0.8\}$, rounded up. Batched stage-3 runs: rounds of 5. Spiked control: 20 genes, 3 hotspots, radius 0.05 of the diagonal, noise 0.1 of the reference standard deviation. Statistics: 5,000 bootstrap resamples; exact sign-flip enumeration over studies; MDE from 300 study resamples with 3,000 sign flips each, on a grid of 51 effect sizes.

The code for the larger slides replaces dense distance computations with sparse and incremental ones; a 132-run check (11 policies, 3 imputers, 2 slide sizes, 2 budgets) reproduced the original code's selections and scores exactly. In the stage-2 run, the five non-adaptive designs reproduced the stage-1 per-slide scores exactly, and the local-variance and diversity policies reproduced their stage-1 means. A caching error interrupted the first stage-2 run. For the results saved before it, a stale cache entry could only have produced a crash, and the error was fixed before any result was analysed.

C Imputer comparison on the earlier slide set

The k-NN and kriging imputers were run only on the earlier slide set. Table 8 reports error-reduction minus the per-slide best non-adaptive design on the five human slides from that run, whose regions held one pseudo-spot each, with the number of slides on which the margin was positive. With five slides these values are descriptive.

Table 8: Error-reduction minus per-slide best non-adaptive design (PCC), mean over 5 human Visium HD slides (positive/total).

Budget	kNN	kriging	graphflow
0.5%	−0.088 (0/5)	−0.088 (2/5)	−0.082 (0/5)
1%	−0.114 (0/5)	−0.138 (1/5)	−0.083 (0/5)
2%	−0.091 (0/5)	−0.150 (1/5)	−0.040 (0/5)
5%	−0.121 (0/5)	−0.141 (1/5)	−0.011 (1/5)
10%	−0.134 (0/5)	−0.133 (0/5)	−0.000 (2/5)
20%	−0.145 (0/5)	−0.126 (0/5)	−0.002 (2/5)

D Stage-1 results

Stage 1 compared policies with the per-slide best design (hindsight). Table 9 gives its 24-margin primary family except the batched and contiguous settings, which are in Table 10; Table 11 gives absolute recall, including the ROI-seeking sampler.

Table 9: Stage-1 primary margins (hindsight benchmarks). Error-reduction is compared with the per-slide best of five non-adaptive designs, local-variance with the per-slide best of k-means, grid, and MI-greedy. Unadjusted 95% CIs resampling studies; p_H Holm-adjusted over the 24 stage-1 primary margins.

	Reconstruction (PCC, top-50 genes)				Detection (ROI recall)	
	error-reduction − best non-adapt.		local-var. − best coverage		local-var. − best coverage	
Budget	Δ [95% CI]	p_H	Δ [95% CI]	p_H	Δ [95% CI]	p_H
0.5%	−0.128 [−0.162, −0.096]	<0.001	–	–	–	–
1%	−0.083 [−0.105, −0.064]	<0.001	–	–	–	–
2%	−0.040 [−0.054, −0.027]	<0.001	−0.136 [−0.171, −0.106]	<0.001	+0.001 [−0.004, +0.006]	0.723
5%	−0.011 [−0.018, −0.004]	0.057	−0.148 [−0.180, −0.119]	<0.001	+0.013 [+0.002, +0.024]	0.362
10%	−0.002 [−0.007, +0.001]	0.687	−0.152 [−0.186, −0.122]	<0.001	+0.021 [+0.003, +0.037]	0.362
20%	−0.002 [−0.004, +0.001]	0.687	−0.165 [−0.217, −0.125]	<0.001	+0.042 [+0.006, +0.077]	0.362

Table 10: Stage-1 local-variance policy minus the per-slide best coverage design (hindsight) at 10% budget under batched and contiguous acquisition. Δ with unadjusted 95% CI resampling studies; p_H Holm-adjusted over the 24 stage-1 primary margins.

	Reconstruction (PCC)		Detection (ROI recall)	
Acquisition setting	Δ [95% CI]	p_H	Δ [95% CI]	p_H
Sequential (1 region per round)	-0.152 $[-0.186, -0.122]$	<0.001	$+0.021$ $[+0.003, +0.037]$	0.362
Rounds of 5 regions	-0.129 $[-0.175, -0.094]$	<0.001	$+0.034$ $[+0.013, +0.053]$	0.073
Rounds of 25 regions	-0.130 $[-0.176, -0.095]$	<0.001	$+0.011$ $[-0.000, +0.021]$	0.362
Single round (whole budget)	-0.236 $[-0.288, -0.188]$	<0.001	-0.051 $[-0.077, -0.027]$	0.008
Contiguous blocks of 7 regions	-0.087 $[-0.116, -0.063]$	<0.001	$+0.024$ $[+0.002, +0.044]$	0.362

Table 11: Stage-1 ROI-detection recall, mean over 22 slides. †Reimplementations of the stated objectives of SCR^2-ST (diversity, non-adaptive) and SOFisher (ROI-seeking, adaptive). *Ground-truth ordering.

	Non-adaptive coverage			Competitor objectives			
Budget	k-means	grid	MI-greedy	diversity†	ROI-seeking†	local-var.	oracle*
2%	0.018	0.022	0.018	0.026	0.021	0.027	0.100
5%	0.050	0.051	0.046	0.067	0.050	0.070	0.250
10%	0.110	0.103	0.094	0.136	0.104	0.137	0.500
20%	0.232	0.204	0.200	0.266	0.232	0.278	0.999

E Feasible comparator choices and the hindsight gap

The feasible procedure selected k-means for reconstruction at 0.5–1% and MI-greedy at 2–20% (with k-means for 4 of 18 studies at 2%), and for detection variance-greedy for most studies at 2–10% and the one-shot ranker at 20% (Table 12).

Table 12: Designs chosen on the other studies (number of held-out studies, of 18) and the difference between the per-slide best design and the feasible design for reconstruction (unadjusted 95% CI resampling studies).

Budget	feasible choice, reconstruction	feasible choice, detection	per-slide best − feasible (PCC)
0.5%	kmeans (18)	–	$+0.021$ $[+0.006, +0.038]$
1%	kmeans (18)	–	$+0.016$ $[+0.006, +0.027]$
2%	mi-greedy (14), kmeans (4)	var-greedy (12), oneshot (5), grid (1)	$+0.017$ $[+0.008, +0.027]$
5%	mi-greedy (18)	var-greedy (18)	$+0.001$ $[+0.000, +0.003]$
10%	mi-greedy (18)	var-greedy (16), oneshot (2)	$+0.001$ $[+0.001, +0.003]$
20%	mi-greedy (18)	oneshot (18)	$+0.002$ $[+0.001, +0.003]$

F Inference for the feasible comparator

The feasible comparator for each held-out study is chosen from the other 17, so the 18 study margins share the selection step, which the study-level tests in the main text do not account for. Two post hoc checks address this (Table 13). The selection-aware bootstrap resamples studies with replacement and repeats the leave-one-study-out choice inside each replicate, excluding copies of the held-out study from its own selection, so its interval reflects uncertainty in which design is chosen. The fixed-comparator analysis replaces the feasible design with MI-greedy on every slide, for which the sign-flip test is valid as specified, with Holm over the 18 analogous margins (the feedback and diversity margins do not involve the comparator and are unchanged). For reconstruction both checks agree with the main analysis. For detection the selection-aware intervals

include zero at every budget, whereas the fixed MI-greedy margins exclude it at 10–20%; MI-greedy has the lowest detection recall in the pool at 5–20%.

Table 13: Stage-2 primary margins against the feasible design with selection-aware 95% bootstrap intervals, and against MI-greedy fixed for every slide (unadjusted 95% CI resampling studies, unadjusted and Holm-adjusted sign-flip p). Post hoc.

	− feasible rule		− MI-greedy (fixed)		
Budget	Δ	selection-aware 95% CI	Δ [95% CI]	p	p_H
Reconstruction (PCC): residual policy					
0.5%	−0.092	[−0.132, −0.020]	−0.055 [−0.094, −0.001]	0.057	0.424
1%	−0.057	[−0.087, −0.013]	−0.045 [−0.072, −0.018]	0.008	0.090
2%	−0.025	[−0.047, −0.011]	−0.033 [−0.047, −0.017]	<0.001	0.014
5%	−0.016	[−0.025, −0.008]	−0.016 [−0.025, −0.009]	<0.001	0.003
10%	−0.007	[−0.011, −0.003]	−0.007 [−0.011, −0.003]	<0.001	0.011
20%	−0.002	[−0.004, −0.000]	−0.002 [−0.004, −0.001]	0.010	0.107
Detection (ROI recall): local-variance policy					
2%	+0.008	[−0.002, +0.013]	+0.009 [+0.003, +0.015]	0.017	0.168
5%	+0.006	[−0.004, +0.022]	+0.023 [+0.010, +0.037]	0.005	0.063
10%	+0.017	[−0.011, +0.041]	+0.043 [+0.020, +0.067]	0.003	0.043
20%	+0.026	[−0.018, +0.085]	+0.078 [+0.036, +0.122]	0.003	0.043

G Stage-2 secondary family

Table 14: Stage-2 secondary margins, 22 slides. Unadjusted 95% CIs resampling studies; p_H Holm-adjusted over the secondary family.

Margin	Budget	Δ [95% CI]	p_H
residual − residual, perm. (PCC)	0.5%	+0.031 [+0.010, +0.057]	0.056
residual − per-slide best (PCC)	0.5%	−0.113 [−0.146, −0.084]	<0.001
residual − residual, perm. (PCC)	1%	+0.039 [+0.021, +0.060]	0.002
residual − per-slide best (PCC)	1%	−0.072 [−0.099, −0.051]	<0.001
residual − residual, perm. (PCC)	2%	+0.030 [+0.014, +0.049]	0.014
residual − per-slide best (PCC)	2%	−0.041 [−0.056, −0.029]	<0.001
local-var. − local-var., constant (recall)	2%	+0.004 [−0.001, +0.010]	0.518
local-var. − feasible (PCC)	2%	−0.119 [−0.156, −0.087]	<0.001
residual − residual, perm. (PCC)	5%	+0.009 [+0.001, +0.018]	0.238
residual − per-slide best (PCC)	5%	−0.018 [−0.026, −0.011]	<0.001
local-var. − local-var., constant (recall)	5%	+0.019 [+0.006, +0.032]	0.083
local-var. − feasible (PCC)	5%	−0.146 [−0.178, −0.117]	<0.001
residual − residual, perm. (PCC)	10%	+0.001 [−0.002, +0.004]	1.000
residual − per-slide best (PCC)	10%	−0.008 [−0.012, −0.005]	<0.001
local-var. − local-var., constant (recall)	10%	+0.034 [+0.011, +0.057]	0.076
local-var. − feasible (PCC)	10%	−0.150 [−0.185, −0.120]	<0.001
residual − residual, perm. (PCC)	20%	−0.001 [−0.003, +0.002]	1.000
residual − per-slide best (PCC)	20%	−0.005 [−0.007, −0.003]	0.003
local-var. − local-var., constant (recall)	20%	+0.074 [+0.032, +0.118]	0.038
local-var. − feasible (PCC)	20%	−0.163 [−0.215, −0.123]	<0.001

H Spiked positive control

Table 15: Semi-synthetic control on the 22 slides: residual policy (and its permuted-feedback version) minus the feasible design, by spike amplitude and budget (PCC; unadjusted 95% CIs resampling studies). The registered success criterion was a positive margin with a CI excluding zero at $\alpha = 2$.

α	Budget	spiked genes in top 50	residual − feasible	residual, perm. − feasible	residual − residual, perm.
0	5%	0	-0.015 $[-0.023, -0.008]$	-0.026 $[-0.034, -0.018]$	$+0.010$ $[+0.002, +0.019]$
0	10%	0	-0.007 $[-0.011, -0.003]$	-0.008 $[-0.012, -0.003]$	$+0.001$ $[-0.002, +0.004]$
1	5%	20	-0.022 $[-0.031, -0.013]$	-0.030 $[-0.038, -0.023]$	$+0.007$ $[-0.001, +0.015]$
1	10%	20	-0.010 $[-0.017, -0.004]$	-0.013 $[-0.019, -0.007]$	$+0.003$ $[-0.002, +0.007]$
2	5%	20	-0.021 $[-0.030, -0.012]$	-0.034 $[-0.043, -0.028]$	$+0.013$ $[+0.004, +0.023]$
2	10%	20	-0.011 $[-0.019, -0.004]$	-0.018 $[-0.026, -0.010]$	$+0.007$ $[-0.000, +0.014]$

I Stage-3 absolute values, secondary family, and permutation seeds

Table 16: Stage-3 mean scores over 22 slides: the fixed coverage design, the hybrids with observed and permuted feedback (two seeds averaged), and the feedback policy run alone for the whole budget (residual policy for reconstruction, local-variance for detection; stage-2 runs). Fixed designs as in Table 5.

		hybrid, 50% coverage		hybrid, 80% coverage		
Budget	fixed rule	observed	permuted	observed	permuted	feedback policy alone
Reconstruction (PCC)						
0.5%	0.449	0.413	0.412	0.441	0.437	0.357
1%	0.534	0.524	0.498	0.525	0.517	0.477
2%	0.599	0.600	0.584	0.605	0.599	0.566
5%	0.659	0.656	0.654	0.658	0.657	0.643
10%	0.685	0.685	0.686	0.686	0.687	0.678
20%	0.707	0.705	0.708	0.706	0.708	0.705
Detection (ROI recall)						
2%	0.023	0.026	0.019	0.024	0.022	0.027
5%	0.064	0.065	0.051	0.063	0.060	0.070
10%	0.127	0.133	0.110	0.128	0.121	0.137
20%	0.252	0.252	0.217	0.249	0.232	0.278

Table 17: Stage-3 secondary margins, 22 slides. Unadjusted 95% CIs resampling studies; p_H Holm-adjusted over the 24 secondary margins. Permuted values are averaged over two seeds.

Margin	Budget	Δ [95% CI]	p_H
reconstruction, hybrid (50% coverage) − permuted	0.5%	+0.0015 [−0.0152, +0.0155]	1.000
reconstruction, hybrid (80% coverage) − permuted	0.5%	+0.0040 [−0.0012, +0.0100]	1.000
reconstruction, hybrid (50% coverage) − permuted	1%	+0.0262 [+0.0065, +0.0540]	0.209
reconstruction, hybrid (80% coverage) − permuted	1%	+0.0078 [−0.0002, +0.0161]	0.749
reconstruction, hybrid (50% coverage) − permuted	2%	+0.0156 [+0.0074, +0.0254]	0.027
detection, hybrid (50% coverage) − permuted	2%	+0.0066 [+0.0039, +0.0093]	0.004
reconstruction, hybrid (80% coverage) − permuted	2%	+0.0067 [+0.0040, +0.0094]	0.016
detection, hybrid (80% coverage) − permuted	2%	+0.0016 [+0.0002, +0.0028]	0.477
reconstruction, hybrid (50% coverage) − permuted	5%	+0.0024 [−0.0013, +0.0062]	1.000
detection, hybrid (50% coverage) − permuted	5%	+0.0141 [+0.0059, +0.0215]	0.113
reconstruction, hybrid (80% coverage) − permuted	5%	+0.0012 [−0.0012, +0.0039]	1.000
detection, hybrid (80% coverage) − permuted	5%	+0.0037 [+0.0011, +0.0059]	0.277
reconstruction, hybrid (50% coverage) − permuted	10%	−0.0008 [−0.0027, +0.0010]	1.000
detection, hybrid (50% coverage) − permuted	10%	+0.0227 [+0.0077, +0.0352]	0.209
reconstruction, hybrid (80% coverage) − permuted	10%	−0.0018 [−0.0031, −0.0006]	0.124
detection, hybrid (80% coverage) − permuted	10%	+0.0074 [−0.0004, +0.0139]	0.749
reconstruction, hybrid (50% coverage) − permuted	20%	−0.0025 [−0.0044, −0.0006]	0.277
detection, hybrid (50% coverage) − permuted	20%	+0.0349 [+0.0097, +0.0570]	0.277
reconstruction, hybrid (80% coverage) − permuted	20%	−0.0019 [−0.0039, −0.0002]	0.477
detection, hybrid (80% coverage) − permuted	20%	+0.0169 [+0.0041, +0.0291]	0.277
rounds of 5: local-variance − variance-greedy	10%	+0.0233 [−0.0001, +0.0463]	0.749
rounds of 5: local-variance − permuted	10%	+0.0497 [+0.0216, +0.0747]	0.113
rounds of 5: local-variance − one-shot	20%	+0.0352 [−0.0168, +0.0952]	1.000
rounds of 5: local-variance − permuted	20%	+0.0839 [+0.0247, +0.1405]	0.240

Table 18: Stage-2 feedback ablations repeated over permutation seeds, 22 slides (descriptive). Seed 2021 is the registered stage-2 run. SD is across the seed means; the seed-averaged margin uses the mean of the permuted runs, with an unadjusted 95% CI resampling studies.

	mean margin by permutation seed						
Budget	2021 (stage 2)	1	2	3	4	SD	seed-averaged Δ [95% CI]
Reconstruction (PCC): residual − permuted							
0.5%	+0.031	+0.024	+0.016	–	–	0.007	+0.023 [+0.008, +0.042]
1%	+0.039	+0.040	+0.030	–	–	0.006	+0.036 [+0.020, +0.055]
2%	+0.030	+0.023	+0.025	–	–	0.004	+0.026 [+0.013, +0.041]
5%	+0.009	+0.010	+0.012	–	–	0.002	+0.010 [+0.002, +0.019]
10%	+0.001	0.000	+0.003	–	–	0.002	+0.001 [−0.002, +0.005]
20%	−0.001	−0.001	+0.001	–	–	0.001	0.000 [−0.002, +0.001]
Detection (ROI recall): local-variance − permuted							
2%	+0.005	+0.003	+0.003	+0.006	+0.007	0.002	+0.005 [−0.000, +0.010]
5%	+0.017	+0.011	+0.016	+0.020	+0.022	0.004	+0.017 [+0.005, +0.030]
10%	+0.029	+0.025	+0.034	+0.035	+0.042	0.006	+0.033 [+0.012, +0.056]
20%	+0.063	+0.075	+0.083	+0.060	+0.090	0.013	+0.074 [+0.031, +0.118]

J Gene-set sensitivity

Table 19: Stage-1 local-variance minus best coverage reconstruction margin (PCC) under three gene sets, 22 slides, unadjusted 95% CIs resampling studies (secondary).

Budget	per-slide top-50 HVG	reference-subset top-50 HVG	all filtered genes
2%	−0.136 [−0.171, −0.106]	−0.134 [−0.168, −0.103]	−0.032 [−0.047, −0.021]
5%	−0.148 [−0.180, −0.119]	−0.148 [−0.182, −0.118]	−0.052 [−0.068, −0.040]
10%	−0.152 [−0.186, −0.122]	−0.153 [−0.189, −0.122]	−0.068 [−0.087, −0.052]
20%	−0.165 [−0.217, −0.125]	−0.166 [−0.217, −0.126]	−0.079 [−0.101, −0.061]

K Platform and tissue splits

The platform split is exploratory and the per-tissue means are descriptive; both use stage-1 margins.

Table 20: Selected stage-1 margins by platform (unadjusted study-level sign-flip p) and by tissue (mean).

	HD (n=7, 5 cl.)	Visium (n=15, 13 cl.)	prostate (n=6)	kidney (n=4)	lung (n=6)	colon (n=6)
Recon., error-red. (1%)	−0.113 (0.062)	−0.068 (<0.001)	−0.078	−0.085	−0.081	−0.088
Recon., local-var. (10%)	−0.103 (0.062)	−0.174 (<0.001)	−0.115	−0.171	−0.141	−0.185
Detect., local-var. (10%)	+0.048 (0.062)	+0.008 (0.426)	−0.001	+0.022	+0.036	+0.027
Detect., local-var. (20%)	+0.099 (0.062)	+0.015 (0.469)	+0.015	+0.024	+0.052	+0.070
Detect., rounds of 5 (10%)	+0.049 (0.250)	+0.026 (0.044)	+0.016	+0.012	+0.046	+0.053

L Native SOFisher run

We ran SOFisher from the authors' repository on its bundled mouse cortex data for 5 million steps (seed 222) with the repository's cortex script. The only change was checkpoint and resume support, so the run could continue after interruptions. Figure 4 shows the evaluation reward, which counts captured target regions. Its median was −238 over the first 5% of training and +147 over the last 10% (best 514). Rare evaluation episodes have rewards near -1.7×10^5, so the figure shows rolling medians.

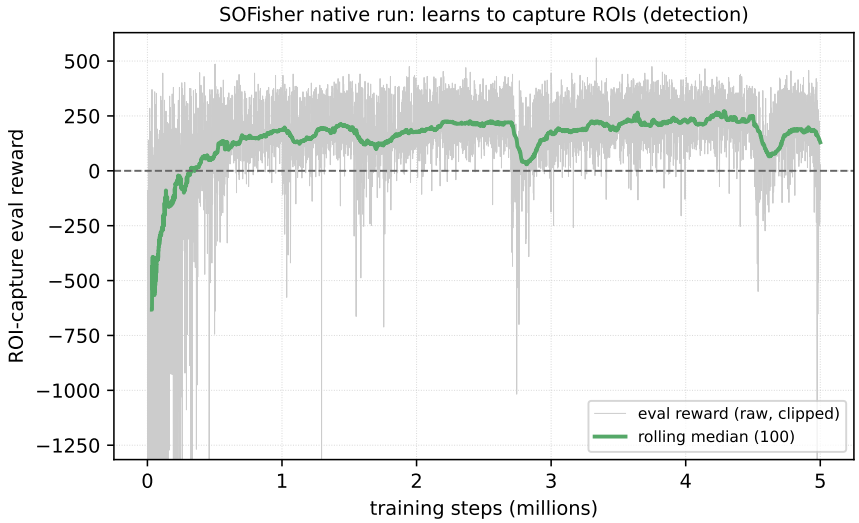


Figure 4: SOFisher evaluation reward during training (rolling median), native run.